

Erdős Problem #790

EP-790	L1 - Tractable	Partiellement résolu	Combinatorics
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Énoncé original (anglais, reproduction mot pour mot)

Let $l(n)$ be maximal such that if $A \subseteq \mathbb{Z}$ with $|A| = n$ then there exists a sum-free $B \subseteq A$ with $|B| \geq l(n)$ - that is, B is such that there are no solutions to $a_1 = a_2 + \dots + a_r$ with $a_i \in B$ all distinct.

Estimate $l(n)$. In particular, is it true that $l(n)n^{-1/2} \rightarrow \infty$? Is it true that $l(n) < n^{1-c}$ for some $c > 0$?

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